

Algorithms and Datastructures — Exercise Sheet 4

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Exercise 1: Hashing with Open Addressing

(a)

- $h_1(23) = 5 \bmod 13 = 5$
- $h_1(12) = 3 \bmod 13 = 3$
- $h_1(75) = 12 \bmod 13 = 12$
- $h_1(945) = 18 \bmod 13 = 5$
 $i = 1: (5 + 1) \bmod 13 = 6$
- $h_1(30) = 3 \bmod 13 = 3$
 $i = 1: (3 + 1) \bmod 13 = 4$
- $h_1(99) = 18 \bmod 13 = 5$
 $i = 2: (5 + 2) \bmod 13 = 7$
- $h_1(345) = 12 \bmod 13 = 12$
 $i = 1: (12 + 1) \bmod 13 = 0$

Index	0	1	2	3	4	5	6	7	8	9	10	11	12
Key	345	-	-	12	30	23	945	99	-	-	-	-	75

(b)

- $h_2(23) = 69 \bmod 13 = 4$
- $h_2(12) = 36 \bmod 13 = 10$
- $h_2(75) = 225 \bmod 13 = 4$
 $(4 + 75 + 1) \bmod 13 = 2$
- $h_2(945) = 2835 \bmod 13 = 1$
- $(h_2(30)) = 90 \bmod 13 = 12$

- $(h_2(99)) = 297 \bmod 13 = 11$
- $(h_2(345)) = 1035 \bmod 13 = 8$

Index	0	1	2	3	4	5	6	7	8	9	10	11	12
Key	-	945	75	-	23	-	-	-	345	-	12	99	30

Exercise 2: Hashing with Chaining

(a)

Assume for contradiction that every bucket has fewer than y elements

Then

$$\sum_{j=0}^{m-1} n_j \leq m \cdot (y - 1) = my - m < my$$

which contradicts $|S| \geq my$.

Therefore the assumption is false

Take $Y = B_j$. Then $|Y| \geq y$ and every element of Y has the same hash value j .

(b)

So the worst-case time for find is proportional to the bucket size, i.e. at least $\Omega(y)$.

In the extreme (all elements in one bucket), find can be $\Theta(n)$ (linear time).

Exercise 3: Application of Hashtables

(a) If the absolute difference between $A(i)$ and $A(j)=A(k)$ exist.

Total operations: $O(n^3)$

(b)

prestore all elements of A in a hash table first, for $O(1)$ lookups.

Check pairwise differences:

- Loop i from 1 to $n-1$ ($O(n)$ times).
- Loop j from 0 to $i-1$ ($O(n)$ times per i).
- Compute $diff = |A[i] - A[j]|$.
- Check if $diff$ is in H ($O(1)$ time per check). If yes, return true.

(c)

Instead of a hash table, sort A first ($O(n \log n)$ time). Then for each (i, j) , compute $diff = |A[i] - A[j]|$ and use binary search ($O(\log n)$ time) to check if $diff$ exists in the sorted A